

ANSWER THE FOLLOWING QUESTIONS.

10*2 1/2 =25

1. Define Vector space.
2. Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and $u = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$ determine if u belongs to null space of A.
3. Find the spanning set of null space of A where $A = \begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & -2 \end{bmatrix}$.
4. Let $b_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $x = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ and $\beta = \{b_1, b_2\}$ then find the coordinate vector $[x]_\beta$ of relative to β .
5. For the subspace $H = \left\{ \begin{bmatrix} s-2t \\ s+t \\ 3t \end{bmatrix} : s, t \in \mathbb{R} \right\}$, find the basis and the dimension.
6. Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigen vector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$ if so, find the eigen value.
7. Find the eigen value of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$.
8. Find the rank of A where $A = \begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix}$.
9. If a 3x8 matrix A has rank 3 find dim Null A, dim row A and Rank A^T .
10. State spanning set theorem.

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EXAM BRANCH

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MAHATMA JYOTHIBA PHULE TELANGANA BACKWARD CLASSES WELFARE RESIDENTIAL DEGREE
COLLEGE(W)

STATION GHANPUR

II YEAR INTERNAL ASSESSMENT-1

SUBJECT: MATHS (DIFFERENTIAL AND VECTOR CALCULUS)

SEM: III

MAX MARKS:25 M

TIME:1 ½ HOUR

ANSWER THE FOLLOWING QUESTIONS.

10*2 1/2 =25

1. Find 1st order partial derivative of $\tan^{-1}(x + y)$.
2. Verify $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$ where $z = \sin^{-1} \frac{x}{y}$
3. State and prove Euler's 1st theorem.
4. Verify Euler's theorem for function $z = \sin^{-1} \frac{x}{y} + \tan^{-1} \frac{y}{x}$.
5. If $u = \sin^{-1} \left[\frac{x^{\frac{1}{3}} + y^{\frac{1}{3}}}{x^{\frac{1}{3}} + y^{\frac{1}{3}} + 1} \right]^{\frac{1}{2}}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{\tan u}{12}$
6. Find $\frac{dz}{dt}$ when $z = xy^2 + x^2y$, $x = at^2$, $y = 2at$ verify by direct substitution.
7. If $H = (y-z, z-x, x-y)$ prove that $\frac{\partial H}{\partial x} + \frac{\partial H}{\partial y} + \frac{\partial H}{\partial z} = 0$.
8. Find $\frac{dy}{dx}$, when $x \sin(x-y) - (x+y) = 0$.
9. Expand the functions $f(x,y) = x^2 + xy - y^2$ by Taylor's theorem power of $(x-1)$ and $(y+2)$.
10. Show that the minimum value of $u = xy + \frac{a^3}{x} + \frac{a^3}{y}$ is $3a^2$

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STATION GHANPUR, MULUGU & MAHABUBABAD
I YEAR
INTERNAL ASSESSMENT – I
SUBJECT: **MATHS(REAL ANALYSIS)**

Semester – II

Max. Marks: 25

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- ❖ Answer all the following questions: $10 \times 2\frac{1}{2} = 25$
1. Define field structure.
 2. Define order structure.
 3. Define neighbourhood of a point and deleted δ -nbd of 'a'.
 4. If M and N are nbds of a point x then show that $M \cap N$ is also a nbd of x.
 5. Find the infimum and supremum of the set $\{\frac{(-1)^n}{n}; n \in \mathbb{N}\}$.
 6. Prove triangle inequalities: For all real numbers x,y. show that i) $|x + y| \leq |x| + |y|$, ii) $|x - y| \geq ||x| - |y||$.
 7. Define sequence with example.
 8. Prove that every convergent sequence is bounded.
 9. Show that every Cauchy sequence is bounded.
 10. Show that $\lim_{n \rightarrow \infty} \frac{3+2\sqrt{n}}{\sqrt{n}} = 2$.

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STATION GHANPUR, MULUGU & MAHABUBABAD
III YEAR

INTERNAL ASSESSMENT – I
SUBJECT: MATHS(ANALYTICAL SOLID GEOMETRY)

Semester – VI

Max. Marks: 25

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- ❖ Answer all the following questions: $10 \times 2\frac{1}{2} = 25$
- Find 't' if the radius of the sphere $x^2 + y^2 + z^2 + 6x - 8y - t = 0$ is 6.
 - Find the center and radius of the sphere $x^2 + y^2 + z^2 - 6x + 2y - 4z + 14 = 0$.
 - Find the equation of the sphere of radius 3 units and concentric with the sphere $x^2 + y^2 + z^2 - 2x - 2y - 2z = 1$.
 - A plane passes through a fixed point (a,b,c) and cuts the axis A,B,C. Show that the locus of the center of the sphere OABC is $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$.
 - Find the equation of a sphere through the circle $x^2 + y^2 + z^2 = 9$, $2x+3y+4z=5$ and point (1,2,3).
 - Find the equation of the sphere with (1,2,3) and (2,3,4) as the ends of the diameter.
 - Define great circle and small circle.
 - Show that the sphere $x^2 + y^2 + z^2 - 25 = 0$, $x^2 + y^2 + z^2 - 24x - 40y - 18z + 225 = 0$ touch externally at the point $(\frac{12}{5}, 4, \frac{9}{5})$.
 - Show that the sphere $x^2 + y^2 + z^2 + 6y + 2z + 8 = 0$, $x^2 + y^2 + z^2 + 6x + 8y + 4z + 20 = 0$ are orthogonal.
 - Define sphere.

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INTERNAL ASSESSMENT-II- 2025-2026

SUBJECT:MATHEMATICS(REAL ANALYSIS)

CLASS:MPCS &MSCS II YEAR

SEMISTER:III

MAX MARKS:25M

TIME:1hr 30m

ANSWER THE FOLLOWING QUESTIONS:

(10X2 ½=25M)

1. Define derivative of a function f at a point 'a'.
2. Prove that if f is differentiable at a point 'a' then f is continuous at 'a'.
3. State and prove Lagrange mean value theorem.
4. Prove that $|\cos x - \cos y| \leq |x - y| \forall x, y \in \mathbb{R}$.
5. Prove that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.
6. Prove that $\lim_{x \rightarrow \infty} x^{\frac{1}{x}} = 1$.
7. Find the Taylor's series for $e^x \forall x \in \mathbb{R}$.
8. If f is continuous at x_0 and g is continuous at $f(x_0)$, then prove that the composite function $g \circ f$ is continuous at x_0 .
9. Let f and g be a real valued function that are continuous at x_0 in $\mathbb{R}$, then prove that
 - i) $f+g$ is continuous at x_0 .
 - ii) fg is continuous at x_0 .
10. Write the Statement of Intermediate value theorem.

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INTERNAL ASSESSMENT-1- 2025-2026

SUBJECT:MATHEMATICS(LINEAR ALGEBRA)

CLASS:MPCS &MECS III YEAR

SEMISTER:05

MAX MARKS:25M

TIME:1hr 30m

ANSWER THE FOLLOWING QUESTIONS:

(10X2 ½=25M)

1. Compute A^8 , where $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$.
2. Diagonalize the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & 5 \end{bmatrix}$.
3. Let $B = \{b_1, b_2, b_3\}$ and $D = \{d_1, d_2\}$ be basis for a vector space V and W respectively .let

$T: V \rightarrow W$ be a linear transformation with the property that $T(b_1) = 3d_1 - 5d_2$,

$T(b_2) = -d_1 + 6d_2$, $T(b_3) = 4d_2$, find the matrix for T relative to B and D.

4. Let $A = \begin{bmatrix} 4 & 3 \\ -3 & 4 \end{bmatrix}$. Find the eigen values of A and find a basis for each eigenspace.
5. Let $A = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$. list the eigen values of A .Give the angle θ of the rotation where
 $-\pi < \theta \leq \pi$ And give the scale factor r.

6. If $u = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $v = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$ then compute $\left(\frac{u \cdot v}{v \cdot v}\right)v$.

7. Show that $\{u_1, u_2, u_3\}$ is an orthogonal set where $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$.

8. Find the orthogonal projection of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ on to the line through $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ and the origin.

9. Find the distance between $u = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$, $z = \begin{bmatrix} -4 \\ -1 \\ 8 \end{bmatrix}$.

10. Find the unit vector in the direction of vector $u = \begin{bmatrix} -30 \\ 40 \end{bmatrix}$.

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INTERNAL ASSESSMENT-1-2025-2026**

SUBJECT : Mathematics(linear algebra)
SEMESTER: V
TIME:1hr:30m

CLASS : MPCS & MECS III YEAR
MAX MARKS: 25

Answer the following questions.

(10x2 ½=25M)

1. Define vector space.
2. Prove that I) $a \cdot 0 = 0 \forall a \in F$
II) $0 \cdot \alpha = 0 \forall \alpha \in V$.
3. If S is a subset of vector space V then $L(S)$ is a subspace of V .
4. Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$, $U = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$ determine if U belongs to null space of A .
5. Find A such that the given set is $W = \left\{ \begin{bmatrix} b-c \\ 2b-3d \\ b+3c-3d \\ c+d \end{bmatrix} \mid b, c, d \in R \right\}$.
6. Show that transformation is linear $T(x, y, z) = (x+y, 0, 2x-z)$.
7. Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigen vector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$ if so find eigen value.
8. Let μ be a eigen value of invertible matrix A . show that μ^{-1} is an eigen value of A^{-1} .
9. Find the rank of A where $A = \begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix}$.
10. If a 3×8 matrix A has rank 3 find $\dim \text{nul } A$, $\dim \text{Row } A$ and $\text{Rank } A^T$.

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STATION GHANPUR@PEMBARTHY
INTERNAL ASSESSMENT-1-2025-2026

SUBJECT : Mathematics(Differential Equations)

CLASS : MPCs & MSCS I YEAR

SEMESTER: I

MAX MARKS: 25

TIME:1hr:30m

Answer the following questions.

(10x 2 ½=25)

1. Solve $\frac{dy}{dx} + \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} = 0$.
2. Solve $\frac{dy}{dx} = \frac{x-y+2}{x+y-1}$.
3. Solve $(2x+2y+3)\frac{dy}{dx} = x+y+1$.
4. Solve $\frac{dy}{dx} = \frac{ax+hy+g}{hx+bg+f}$.
5. Solve $(x^2 - y^2)dx + 2xy dy = 0$.
6. Solve $3e^x \tan y dx + (1-e^x) \sec^2 y dy = 0$.
7. Solve $(1+e^{\frac{x}{y}})dx + e^{\frac{x}{y}}(1-\frac{x}{y})dy = 0$.
8. Solve $(1+y^2)dx = (\tan^{-1}y - x)dy$.
9. Solve $(1-x^2)\frac{dy}{dx} + 2xy = x(1-x^2)^{1/2}$.
10. Solve $(e^y + 1)\cos x dx + e^y \sin x dy = 0$.

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INTERNAL ASSESSMENT-1-2025-2026**

SUBJECT : Mathematics(Real analysis)
SEMESTER: III
TIME:1hr:30m

CLASS : MPCS & MSCS II YEAR
MAX MARKS: 25

Answer the following questions.

(10 x2 ½=25 M)

1. Define convergent sequence. Find Limit of $S_n = 3 + \frac{1}{n}$.
2. If $\lim S_n = 0$ then prove that $\lim |S_n| = 0$.
3. State and prove sandwich theorem.
4. Every convergent sequence is a Cauchy sequence.
5. If $\sum_{n=1}^{\infty} U_n$ converges then $\lim_{n \rightarrow \infty} U_n = 0$.
6. Write conditions for Geometric series, P-test, comparison test.
7. Test the convergence of the series a) $\sum_{n=0}^{\infty} \frac{n}{2^n}$ b) $\sum_{n=0}^{\infty} \frac{n^2}{n!}$.
8. Define lower Riemann sum and upper Riemann sum.
9. Let f be a bounded function on $[a, b]$ then $L(f) \leq U(f)$.
10. Let $f(x) = 1$ for rational x in $[a, b]$ and let $f(x) = 0$ for irrational x in $[0, b]$. Is f integrable on $[0, b]$.

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